MudWatt Bioelectric League

Turn a cup of mud into a living battery

At a glance: Can your team make mud generate electricity and defend your design with data? Plan about 80 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: microbial fuel cells, electrons, decomposition, bioenergy (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- carbon felt electrodes
- 24 oz deli cups
- 100 kohm resistors
- alligator leads
- multimeters, one per team
- sugar cubes for fuel
- fresh soil or mud samples
- nitrile gloves
- labels and masking tape
- handwashing supplies

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

01 Set up one station per team with the materials above, sorted and ready.
02 Stage the scoring sheet and any reference values before testing begins.
03 Put out PPE (gloves, goggles, trays) before any materials are handled.
04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Can your team make mud generate electricity and defend your design with data?" Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints

0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Build or solve

0:18 - 0:44

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Test and record

Teams run the standard test and log a baseline result, the voltage in DC millivolts across the 100 kohm load, before changing anything.

Phase 6 Redesign

Each team changes exactly one variable, predicts the effect, and re-tests to compare against baseline.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, and reset the station. PPE off, wash or wipe down.

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: No measurable result on the first test. **Fix:** Check the setup against the steps, confirm the standard test was followed, and log the baseline anyway.

Problem: Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to electrogenic bacteria.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Which design change moved your readings the most, and how do you know?
- Why must the anode stay buried and the cathode stay in air?
- How is your weekly log like the work of an energy engineer?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Peak voltage (mV)	35
Daily data log quality	20
Controlled-variable design	15
Redesign or maintenance plan	15
Final explanation	15
Total	**100**

CLEANUP AND SOURCES

Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

SOURCES Ohio State Microbial Fuel Cell Learning Center